

Boss Challenge — Paper 35 (Class 7)

Nutrition in Plants | Respiration in Organisms | Lines & Angles | Perimeter & Area

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The process by which green plants prepare their own food using sunlight, water and air is called:
(A) respiration
(B) transpiration
(C) digestion
(D) photosynthesis
2. The process in which living cells break down food to release energy is called:
(A) photosynthesis
(B) digestion
(C) excretion
(D) respiration
3. An angle that measures exactly 90° , like the corner of a book, is called a:
(A) right angle
(B) straight angle
(C) obtuse angle
(D) acute angle
4. A gardener wants to put a fence right along the outer edge of a rectangular plot; the fencing length needed equals the plot's:
(A) perimeter
(B) diagonal
(C) area
(D) volume
5. Green plants are described as autotrophs mainly because they:
(A) eat other animals
(B) depend fully on other plants
(C) make their own food
(D) never need any food at all
6. During normal breathing, the gas that humans take IN from the air and use is:
(A) hydrogen
(B) nitrogen
(C) oxygen
(D) carbon dioxide
7. An angle whose measure is less than 90° is called an:
(A) right angle
(B) reflex angle
(C) obtuse angle
(D) acute angle

8. To work out how many floor tiles are needed to cover a room's floor completely, you must first find the floor's:
- (A) volume
 - (B) area
 - (C) perimeter
 - (D) height
9. The green colouring substance in leaves that traps sunlight for food-making is called:
- (A) starch
 - (B) chlorine
 - (C) haemoglobin
 - (D) chlorophyll
10. The gas that humans breathe OUT in larger amounts than they breathe in is:
- (A) hydrogen
 - (B) oxygen
 - (C) nitrogen
 - (D) carbon dioxide
11. An angle that is greater than 90° but less than 180° is called an:
- (A) obtuse angle
 - (B) acute angle
 - (C) right angle
 - (D) straight angle
12. The perimeter of a square whose side is 5 cm long, found by $4 \times \text{side}$, is:
- (A) 10 cm
 - (B) 25 cm
 - (C) 20 cm
 - (D) 9 cm
13. The tiny pores on the surface of a leaf through which gases pass in and out are called:
- (A) roots
 - (B) veins
 - (C) stomata
 - (D) petals
14. Respiration that uses oxygen to break down food fully is called:
- (A) aerobic respiration
 - (B) anaerobic respiration
 - (C) transpiration
 - (D) photosynthesis

15. An angle of exactly 180° , where the two arms form one straight line, is called a:
- (A) reflex angle
 - (B) right angle
 - (C) straight angle
 - (D) complete angle
16. The area of a square whose side is 5 cm long, found by side x side, is:
- (A) 15 cm^2
 - (B) 10 cm^2
 - (C) 20 cm^2
 - (D) 25 cm^2
17. Besides chlorophyll and sunlight, the two raw materials a leaf needs to make food are water and:
- (A) smoke
 - (B) oxygen
 - (C) carbon dioxide
 - (D) nitrogen
18. Respiration that takes place without using oxygen is called:
- (A) aerobic respiration
 - (B) anaerobic respiration
 - (C) photosynthesis
 - (D) germination
19. An angle of 35° is placed beside an angle of 55° ; because their measures total 90° , this pair is called:
- (A) complementary angles
 - (B) adjacent angles
 - (C) vertically opposite angles
 - (D) supplementary angles
20. The perimeter of a rectangle is correctly found using the formula:
- (A) $2 \times (\text{length} + \text{breadth})$
 - (B) $4 \times \text{length}$
 - (C) $\text{length} \times \text{breadth}$
 - (D) $\text{length} + \text{breadth}$ alone
21. The food first made by green leaves during photosynthesis is a simple sugar called:
- (A) fat
 - (B) protein
 - (C) glucose
 - (D) vitamin

22. When our muscles run short of oxygen during hard exercise, the food is broken down to give energy and:

- (A) starch
- (B) lactic acid
- (C) alcohol
- (D) pure oxygen

23. Angles of 110° and 70° lie next to each other along a straight line and total 180° ; such a pair is called:

- (A) supplementary angles
- (B) corresponding angles
- (C) vertically opposite angles
- (D) complementary angles

24. The area of a rectangle is correctly found using the formula:

- (A) length x breadth
- (B) length + breadth alone
- (C) $2 \times (\text{length} + \text{breadth})$
- (D) side x side

25. During photosynthesis, the gas that green plants take IN from the surrounding air is:

- (A) nitrogen
- (B) hydrogen
- (C) carbon dioxide
- (D) oxygen

26. The painful muscle cramps felt after running very fast are caused by a build-up of:

- (A) undigested food
- (B) lactic acid in the muscles
- (C) extra oxygen
- (D) water in the bones

27. If an angle measures 30° , then its complement, the angle that completes it to 90° , is:

- (A) 150°
- (B) 60°
- (C) 33°
- (D) 70°

28. A rectangle is 7 cm long and 4 cm wide. Its perimeter, found by $2 \times (7 + 4)$, is:

- (A) 28 cm
- (B) 22 cm
- (C) 11 cm
- (D) 14 cm

- 29.** During photosynthesis, the gas that green plants give OUT into the air is:
- (A) carbon dioxide
 - (B) nitrogen
 - (C) oxygen
 - (D) water vapour
- 30.** In yeast, anaerobic respiration breaks down sugar to give alcohol, energy and the gas:
- (A) oxygen
 - (B) hydrogen
 - (C) nitrogen
 - (D) carbon dioxide
- 31.** If an angle measures 110° , then its supplement, the angle that completes it to 180° , is:
- (A) 250°
 - (B) 70°
 - (C) 80°
 - (D) 90°
- 32.** A rectangle is 7 cm long and 4 cm wide. Its area, found by 7×4 , is:
- (A) 11 cm^2
 - (B) 22 cm^2
 - (C) 14 cm^2
 - (D) 28 cm^2
- 33.** To test whether a leaf has made food, a few drops of this solution are dripped onto it:
- (A) salt water, which turns clear
 - (B) iodine, which turns blue-black with starch
 - (C) litmus, which turns red
 - (D) lime water, which turns milky
- 34.** In a living organism, the breakdown of food to release energy takes place mainly inside the:
- (A) hair
 - (B) nails
 - (C) cells
 - (D) bones
- 35.** Two angles that share a common vertex and a common arm and lie on opposite sides of that arm are called:
- (A) adjacent angles
 - (B) vertically opposite angles
 - (C) alternate angles
 - (D) complementary angles

36. The area of a right-angled triangle with base 6 cm and height 4 cm, found by $\frac{1}{2} \times \text{base} \times \text{height}$, is:
- (A) 12 cm²
 - (B) 24 cm²
 - (C) 10 cm²
 - (D) 20 cm²
37. Living things that cannot make their own food and depend on others for it are called:
- (A) autotrophs
 - (B) producers
 - (C) heterotrophs
 - (D) chlorophylls
38. The taking in of air rich in oxygen into the body is called:
- (A) inhalation
 - (B) respiration
 - (C) digestion
 - (D) exhalation
39. A pair of adjacent angles whose outer (non-common) arms together form a straight line is called a:
- (A) vertical pair
 - (B) linear pair
 - (C) equal pair
 - (D) complementary pair
40. The area of a triangle is correctly found using the formula:
- (A) base x height alone
 - (B) length x breadth
 - (C) side x side
 - (D) $\frac{1}{2} \times \text{base} \times \text{height}$
41. Cuscuta (amarbel), a pale climber that draws its food from another living plant, is a:
- (A) saprotroph
 - (B) parasite
 - (C) insectivore
 - (D) autotroph
42. The giving out of air rich in carbon dioxide from the body is called:
- (A) inhalation
 - (B) transpiration
 - (C) exhalation
 - (D) respiration

43. The two angles that together form a linear pair always add up to:
- (A) 100°
 - (B) 180°
 - (C) 90°
 - (D) 360°
44. The area of a parallelogram is correctly found using the formula:
- (A) $2 \times (\text{base} + \text{height})$
 - (B) $\text{side} \times \text{side}$
 - (C) $\text{base} \times \text{height}$
 - (D) $\frac{1}{2} \times \text{base} \times \text{height}$
45. The pitcher plant traps and digests insects mainly because the soil it grows in is poor in:
- (A) carbon dioxide
 - (B) nitrogen
 - (C) water
 - (D) sunlight
46. One complete inhalation taken together with one exhalation makes up one:
- (A) step
 - (B) pulse
 - (C) breath
 - (D) heartbeat
47. When two straight lines cross, the pair of angles lying directly opposite each other are equal and are called:
- (A) adjacent angles
 - (B) vertically opposite angles
 - (C) corresponding angles
 - (D) complementary angles
48. A parallelogram has a base of 10 cm and a height of 4 cm. Its area, found by 10×4 , is:
- (A) 14 cm^2
 - (B) 40 cm^2
 - (C) 28 cm^2
 - (D) 20 cm^2
49. A plant such as the pitcher plant that catches and feeds on insects is called an:
- (A) saprotrophic plant
 - (B) parasitic plant
 - (C) aquatic plant
 - (D) insectivorous plant

50. A boy at rest counts 18 complete breaths in one minute; this 18-per-minute figure for him is his:
- (A) pulse rate
 - (B) growth rate
 - (C) breathing rate
 - (D) heart rate
51. The two long rails of a railway track keep the same gap apart and never cross, no matter how far they run; they are:
- (A) perpendicular lines
 - (B) intersecting lines
 - (C) parallel lines
 - (D) curved lines
52. One square metre (1 m^2) is equal to this many square centimetres:
- (A) 1000000 cm^2
 - (B) 1000 cm^2
 - (C) 10000 cm^2
 - (D) 100 cm^2
53. Mushrooms and bread mould obtain their food from dead and decaying matter, so they are called:
- (A) saprotrophs
 - (B) insectivores
 - (C) parasites
 - (D) autotrophs
54. The large dome-shaped muscle lying just below the lungs that helps in breathing is the:
- (A) heart
 - (B) stomach
 - (C) diaphragm
 - (D) liver
55. Two lines that cross each other forming an angle of exactly 90° are said to be:
- (A) acute to each other
 - (B) perpendicular
 - (C) parallel
 - (D) equal
56. The distance once around a square park whose side is 25 m, found by $4 \times \text{side}$, is:
- (A) 625 m
 - (B) 100 m
 - (C) 50 m
 - (D) 75 m

57. The close partnership in which a fungus and an alga live together and help each other is called a:
- (A) saprotroph
 - (B) lichen
 - (C) root nodule
 - (D) parasite
58. In humans, the exchange of gases takes place in the tiny balloon-like air sacs of the lungs called:
- (A) villi
 - (B) nephrons
 - (C) neurons
 - (D) alveoli
59. A line that cuts across two or more other lines at separate points is called a:
- (A) bisector
 - (B) transversal
 - (C) perpendicular line
 - (D) parallel line
60. The area of a square field whose side is 20 m, found by side x side, is:
- (A) 40 m²
 - (B) 200 m²
 - (C) 80 m²
 - (D) 400 m²
61. Farmers often grow leguminous plants such as beans and peas because their roots help to enrich the soil with:
- (A) oxygen
 - (B) carbon
 - (C) plain water
 - (D) nitrogen
62. Insects take in air for respiration through tiny openings along the sides of their body called:
- (A) gills
 - (B) nostrils
 - (C) spiracles
 - (D) stomata
63. When a transversal cuts two parallel lines, each pair of corresponding angles turns out to be:
- (A) always exactly 90°
 - (B) always exactly 45°
 - (C) supplementary, adding to 180°
 - (D) equal

64. The proper kind of unit used to measure the AREA of a surface is a:
- (A) square unit, such as cm^2 or m^2
 - (B) unit of time, such as seconds
 - (C) cubic unit, such as cm^3
 - (D) plain length unit, such as cm
65. The bacteria living in the root nodules of pulse plants are useful chiefly because they can fix:
- (A) oxygen from the air
 - (B) nitrogen from the air
 - (C) sunlight for food
 - (D) water from the soil
66. Fish are able to breathe under water using special organs that take oxygen from the water, called:
- (A) gills
 - (B) lungs
 - (C) nostrils
 - (D) spiracles
67. When a transversal crosses two parallel lines, a pair of alternate interior angles turns out to be:
- (A) always right angles
 - (B) supplementary, adding to 180°
 - (C) always reflex angles
 - (D) equal
68. The proper kind of unit used to measure the PERIMETER of a figure is a:
- (A) unit of length, such as cm or m
 - (B) square unit, such as cm^2
 - (C) unit of mass, such as grams
 - (D) cubic unit, such as cm^3
69. Water and dissolved minerals are taken up from the soil into a plant chiefly by its:
- (A) roots
 - (B) leaves
 - (C) fruits
 - (D) flowers
70. An earthworm has no lungs or gills, so it breathes mainly through its moist:
- (A) gills
 - (B) skin
 - (C) lungs
 - (D) spiracles

71. The angle between the hour hand and the minute hand of a clock at exactly 3 o'clock is:
- (A) 360° , a complete turn
 - (B) 90° , a right angle
 - (C) 45° , half a right angle
 - (D) 180° , a straight angle
72. To work out how much fencing is needed to go right around a field, you must first find its:
- (A) perimeter
 - (B) area
 - (C) diagonal
 - (D) volume
73. A leaf is often called the 'kitchen' or food factory of a green plant because it:
- (A) stores all the plant's seeds
 - (B) anchors the plant in the soil
 - (C) carries water up to the flowers
 - (D) carries out photosynthesis to make food
74. Like animals, green plants also respire, exchanging respiratory gases through the leaf pores called:
- (A) spiracles
 - (B) veins
 - (C) alveoli
 - (D) stomata
75. If the two angles of a linear pair are equal to each other, then each of them must measure:
- (A) 90°
 - (B) 60°
 - (C) 180°
 - (D) 45°
76. To work out how many tiles are needed to cover a floor, you must first find its:
- (A) diagonal
 - (B) area
 - (C) perimeter
 - (D) height
77. Photosynthesis in a green plant can take place only when there is:
- (A) moonlight only
 - (B) soil but no light at all
 - (C) total darkness
 - (D) sunlight

- 78.** The roots of a plant get the oxygen they need for respiration from air present in the:
- (A) water deep underground
 - (B) middle of solid rocks
 - (C) spaces between the soil particles
 - (D) leaves far above them
- 79.** An angle of exactly 360° , made by turning all the way round back to the start, is called a:
- (A) right angle
 - (B) straight angle
 - (C) complete angle
 - (D) reflex angle
- 80.** A square and a rectangle have the SAME perimeter. For the same perimeter, the shape with the larger area is usually the:
- (A) square
 - (B) they are always equal
 - (C) it cannot be decided at all
 - (D) long thin rectangle
- 81.** Apart from the leaves, photosynthesis can also take place in a plant's stem when the stem is:
- (A) very thick and tall
 - (B) brown and woody
 - (C) buried underground
 - (D) green (contains chlorophyll)
- 82.** We pant and breathe much faster after running because the body suddenly needs more:
- (A) food
 - (B) oxygen
 - (C) carbon dioxide
 - (D) water
- 83.** Two complementary angles happen to be equal to each other. Each of these angles therefore measures:
- (A) 90°
 - (B) 60°
 - (C) 30°
 - (D) 45°
- 84.** If the side of a square is doubled, while it stays a square, its area becomes:
- (A) two times as large
 - (B) four times as large
 - (C) the same as before
 - (D) eight times as large

85. Growing the same crop again and again drains the soil of nutrients, which farmers put back by adding:

- (A) manure or fertilisers
- (B) extra sunlight
- (C) plain dry sand
- (D) more stones

86. A simple way to show that exhaled air carries more carbon dioxide is that, blown through it, lime water turns:

- (A) clear and colourless
- (B) bright red
- (C) deep blue
- (D) milky

87. An angle whose measure is more than 180° but less than 360° is called a:

- (A) obtuse angle
- (B) complete angle
- (C) reflex angle
- (D) straight angle

88. A rectangle measures 12 cm by 5 cm. Its area, found by 12×5 , is:

- (A) 24 cm^2
- (B) 17 cm^2
- (C) 34 cm^2
- (D) 60 cm^2

89. Inside the green cells of a leaf, chlorophyll is held in tiny green bodies called:

- (A) chloroplasts
- (B) stomata
- (C) chromosomes
- (D) nuclei

90. A resting person breathes 15 times each minute. The number of breaths taken in 4 minutes, found by 15×4 , is:

- (A) 60 breaths
- (B) 11 breaths
- (C) 45 breaths
- (D) 19 breaths

91. When two lines cross, one of the angles formed measures 65° . The angle vertically opposite to it measures:

- (A) 90°
- (B) 115°
- (C) 65°
- (D) 25°

- 92.** An equilateral triangle has all three sides 9 cm long. Its perimeter, found by 3×9 , is:
- (A) 27 cm
 - (B) 18 cm
 - (C) 81 cm
 - (D) 12 cm
- 93.** A square leaf has each side 6 cm long. The flat surface over which it can catch sunlight is, with $\text{area} = \text{side} \times \text{side}$:
- (A) 18 cm^2
 - (B) 12 cm^2
 - (C) 24 cm^2
 - (D) 36 cm^2
- 94.** An athlete's breathing rate rises from 18 breaths per minute at rest to 30 during a run. The increase, found by $30 - 18$, is:
- (A) 12 breaths per minute
 - (B) 30 breaths per minute
 - (C) 2 breaths per minute
 - (D) 48 breaths per minute
- 95.** One angle of a linear pair measures 120° . The other angle of that pair measures:
- (A) 120°
 - (B) 60°
 - (C) 90°
 - (D) 240°
- 96.** A rectangular vegetable garden is 10 m long and 6 m wide. The fence needed to enclose it, found by $2 \times (10 + 6)$, is:
- (A) 16 m
 - (B) 60 m
 - (C) 32 m
 - (D) 20 m
- 97.** A rectangular leaf measures 8 cm in length and 3 cm in breadth. The surface it spreads out to catch sunlight is:
- (A) 11 cm^2
 - (B) 22 cm^2
 - (C) 16 cm^2
 - (D) 24 cm^2
- 98.** During daytime a green plant both photosynthesises and respire, but overall it gives out more:
- (A) nitrogen than oxygen
 - (B) carbon dioxide than oxygen
 - (C) oxygen than carbon dioxide
 - (D) water than oxygen

99. At exactly 6 o'clock, the hour and minute hands point in opposite directions, forming an angle of:

- (A) 90° , a right angle
- (B) 60° , an acute angle
- (C) 360° , a complete turn
- (D) 180° , a straight angle

100. A square seed-bed of side 4 m is to be covered with a layer of soil. The area to be covered, found by 4×4 , is:

- (A) 16 m^2
- (B) 12 m^2
- (C) 8 m^2
- (D) 16 m